

Arduino Microcontroller Concepts

CSE 4820





ABOUT YOUR SUBJECT

Subject Name : [Arduino Microcontroller Concepts](#)

Code : [CSE 4820](#)

Credits : [4](#)

Grading Pattern : [1](#)

Contact Hours : [3L-2P](#)

Textbook-1 : Textbooks – [Arduino Cookbook — Recipes to Begin, Expand, and Enhance Your Projects](#) by Margolis, Shroff / O'Reilly





WHAT STUDENTS WILL LEARN?

- Gain a strong foundation in **Arduino hardware and IDE programming**.
- Develop the ability to **connect and interface sensors, actuators, and modules** with Arduino.
- Learn how to **apply programming logic** for controlling motors, sensors, etc.
- Strengthen **debugging and troubleshooting skills** in hardware–software integration.
- Improve **teamwork, collaboration, and project presentation skills**.



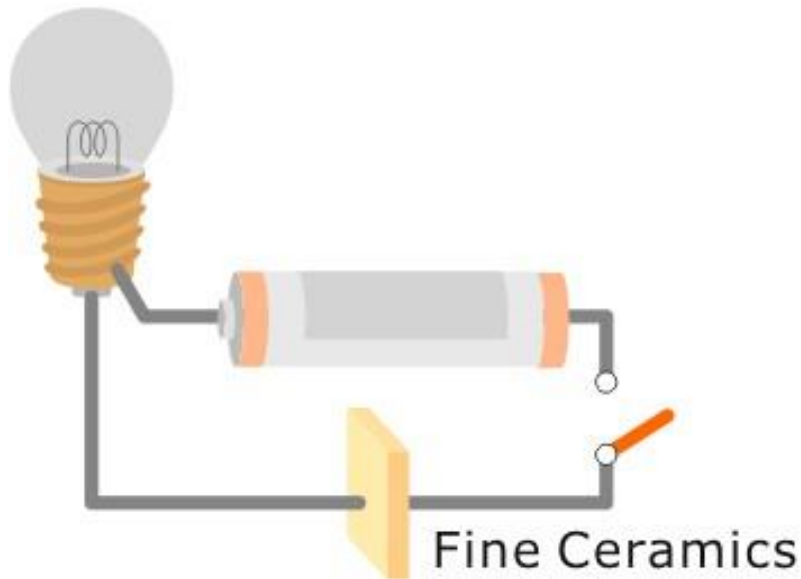


BASIC ELECTRONICS

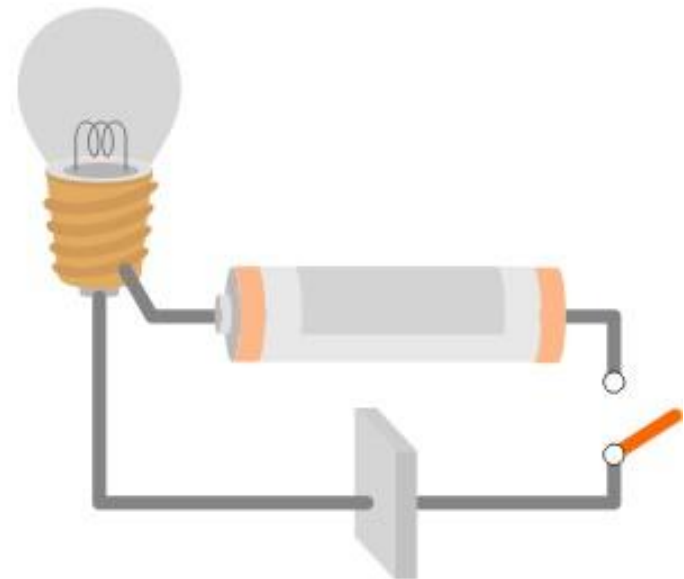


ELECTRICAL INSULATOR & CONDUCTOR:

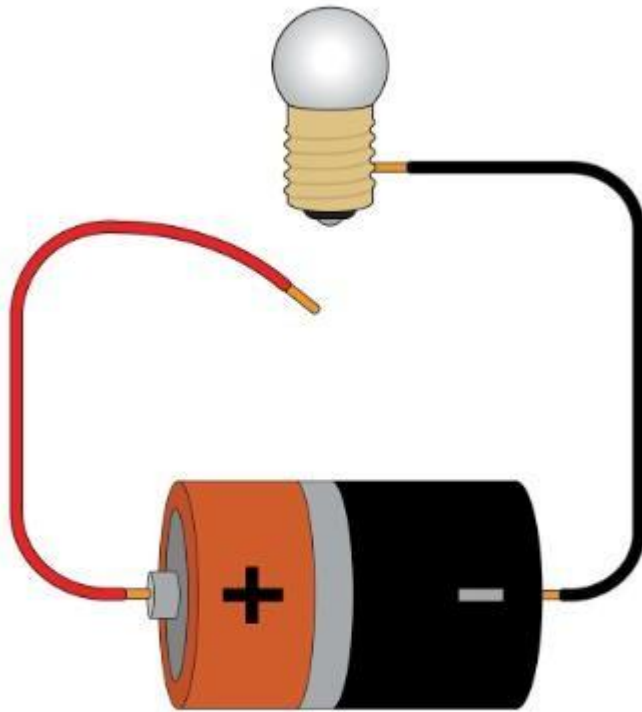
Insulating Ceramics



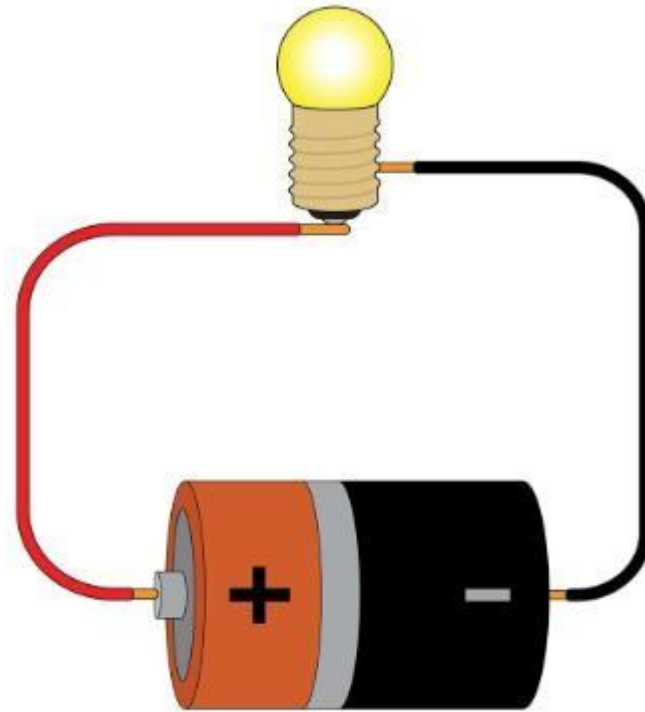
Metals



OPEN AND CLOSE CIRCUIT:



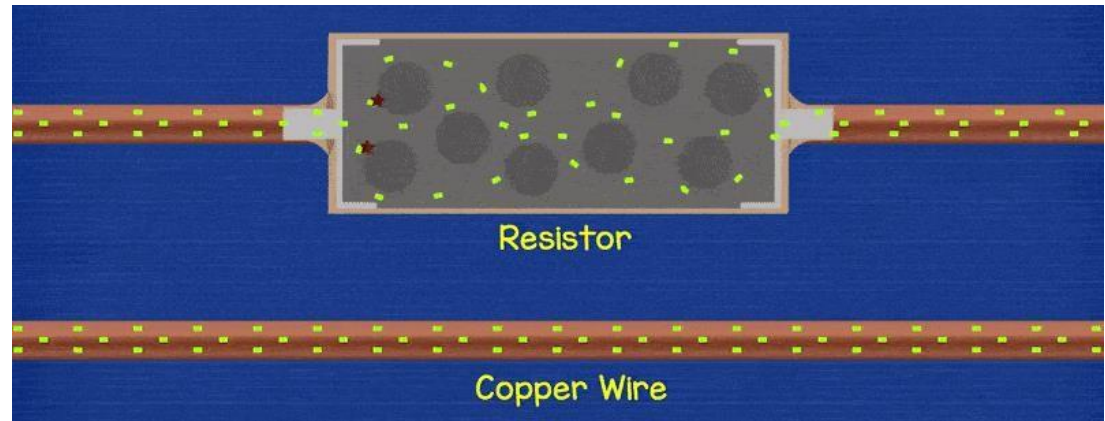
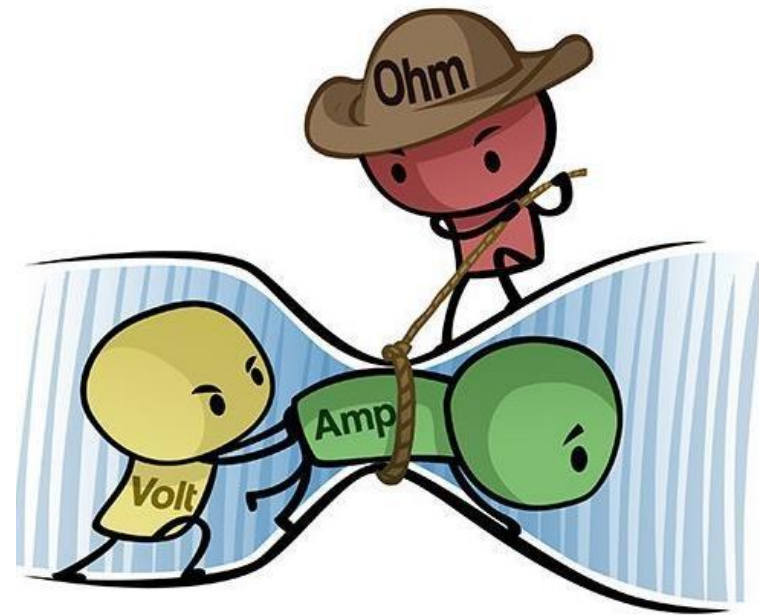
Open Circuit



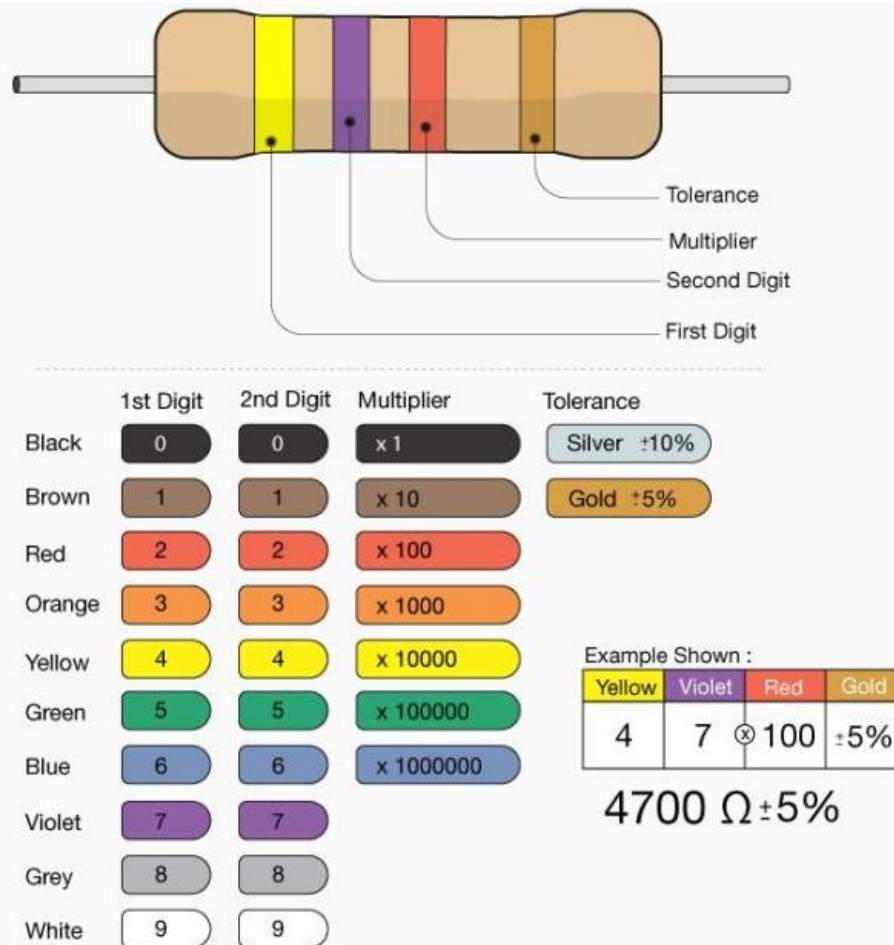
Closed Circuit



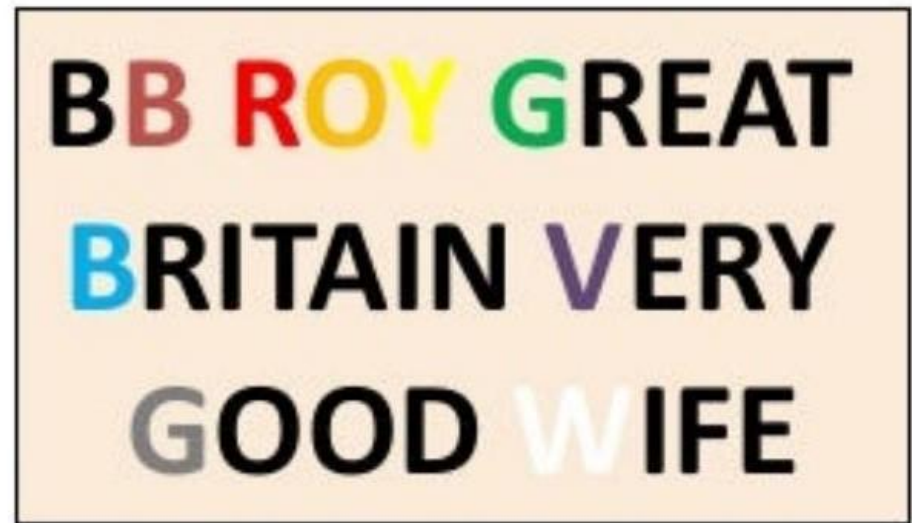
RESISTOR



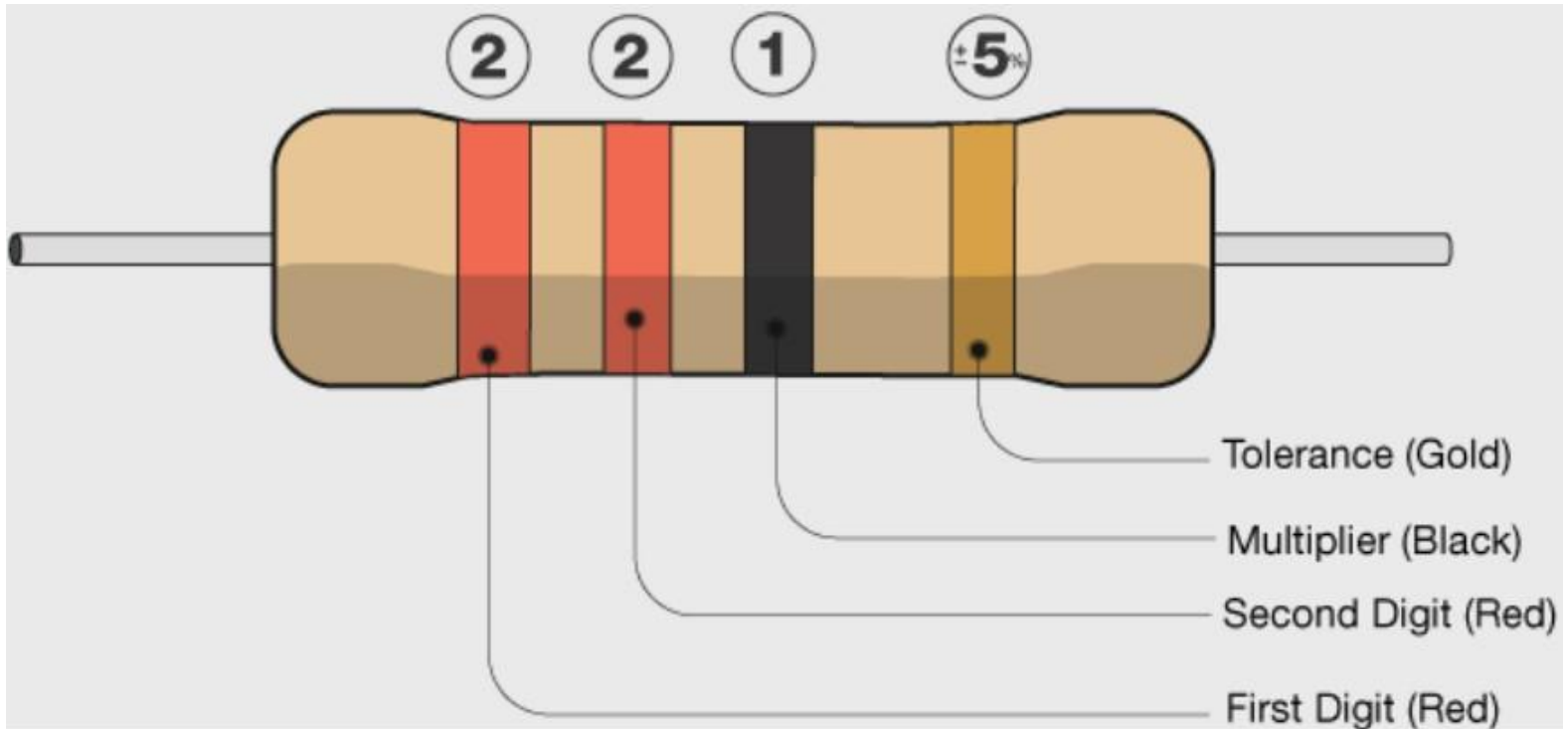
RESISTOR COLOR CODE:



Mnemonic



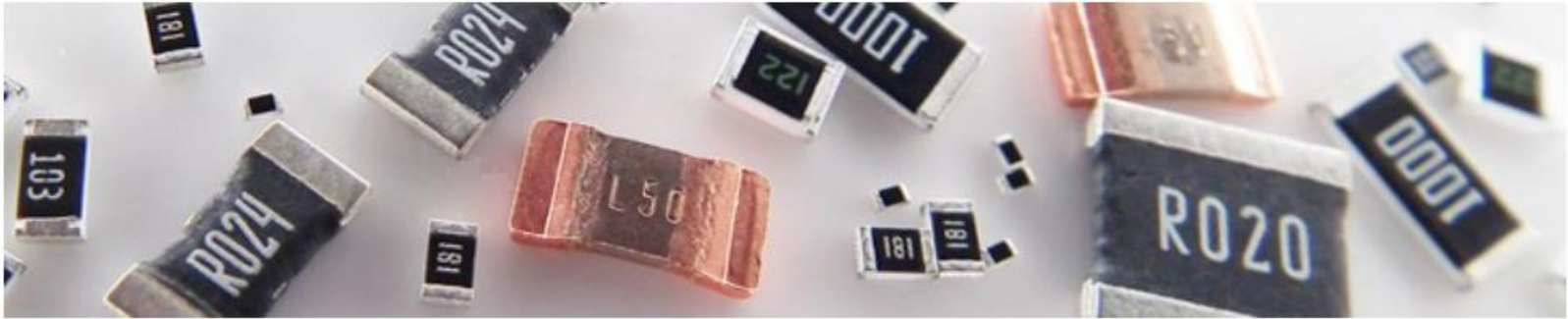
COLOR CODE CALCULATION:



Band colours in order	RED	RED	BLACK	GOLD
Digit representation	2	2	1	±5%
Value	22 Ω ±5 %			



SURFACE MOUNT RESISTOR CALCULATION:



102 → $10 \times 10^2 = 1000\Omega$ (1k Ω)

1002 → $100 \times 10^2 = 10000\Omega$ (10 k Ω)

R047 → Read "R" as a decimal point. → 0.047 Ω (47m Ω)

0K47 → Read "K" as a decimal point. → 0.47K Ω

4M7 → Read "M" as a decimal point. → 4.7M Ω

10L0 → Read "L" as a decimal point of m Ω → 10m Ω



TYPES OF RESISTOR:



Fixed Resistor



Thick Film Type



LDR Resistor



Thin Film Type



Potentiometer



Trimming Pot



Surface Mount



Thermistor



Rheostat



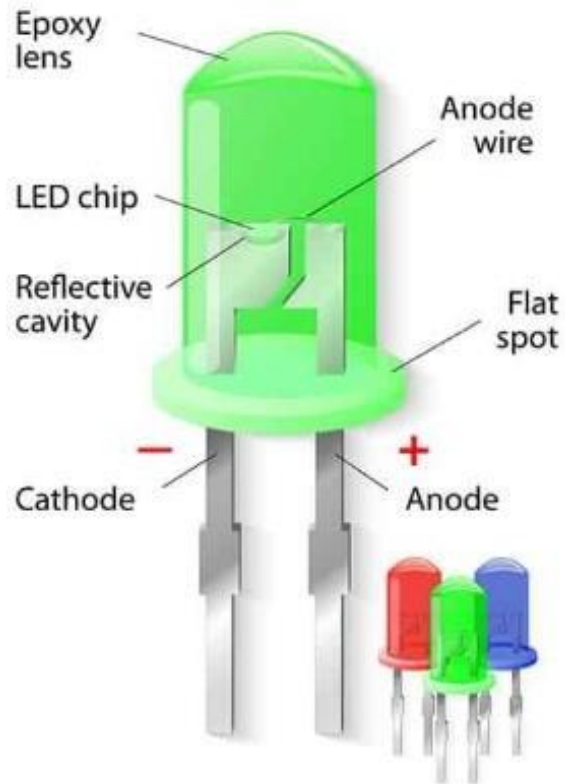
Varistor



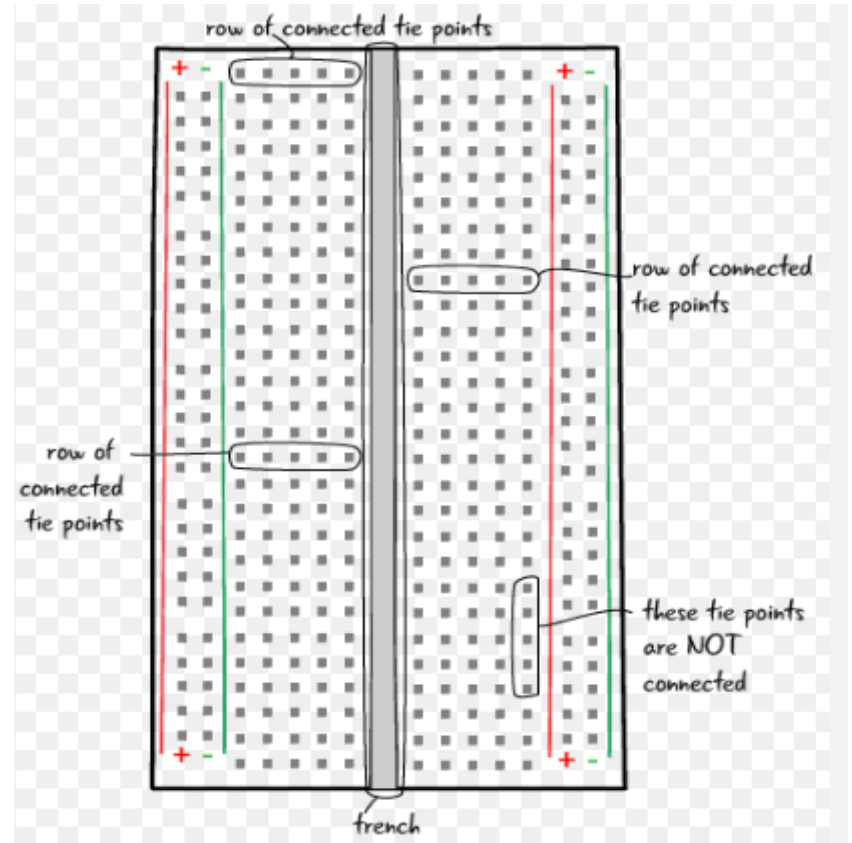
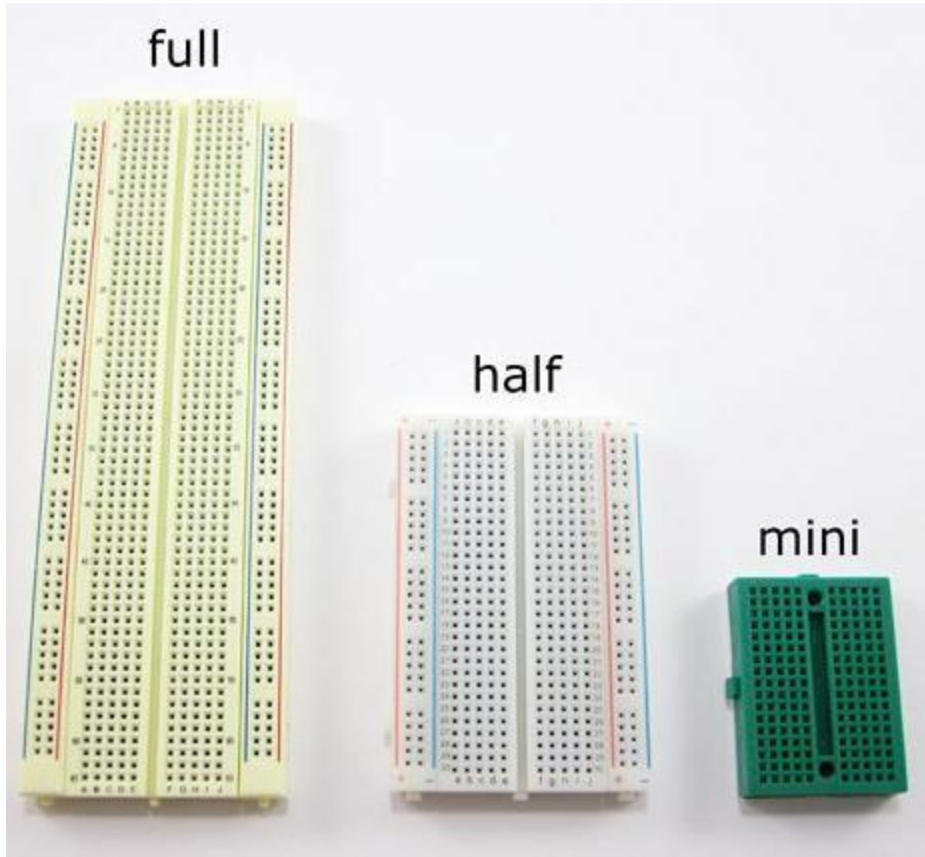
Wire Wound Type



LIGHT EMITTING DIODE (LED):

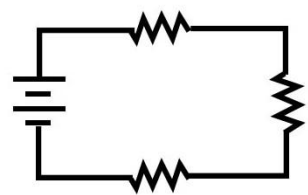


BREAD BOARD:

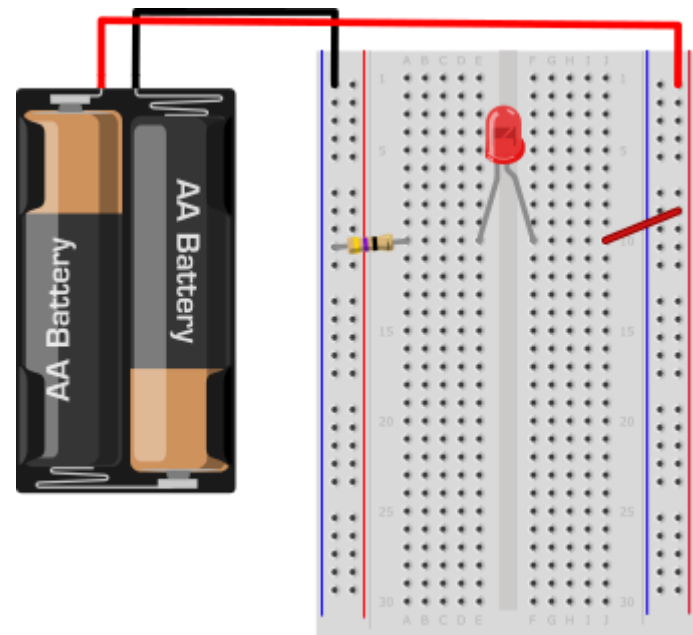
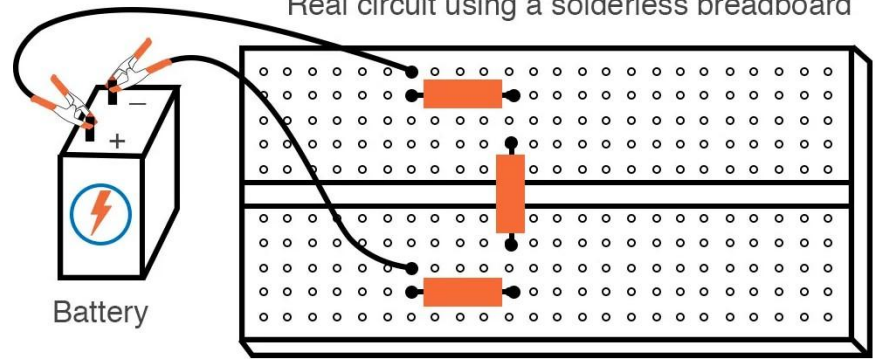


SCHEMATIC TO BREAD BOARD IMPLEMENTATION:

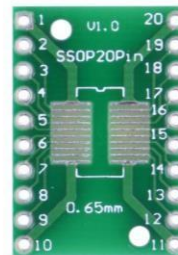
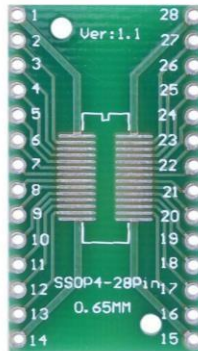
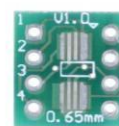
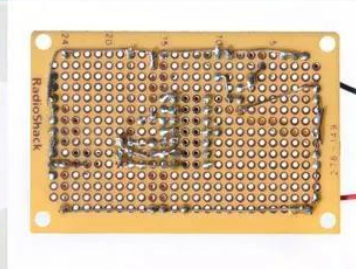
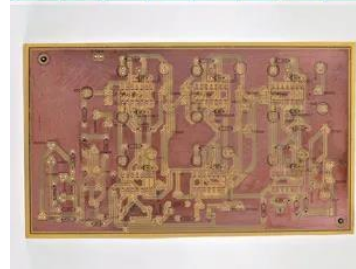
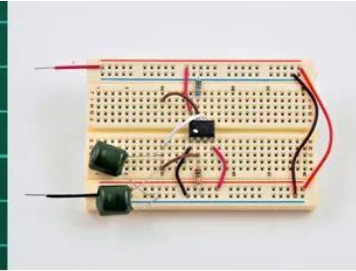
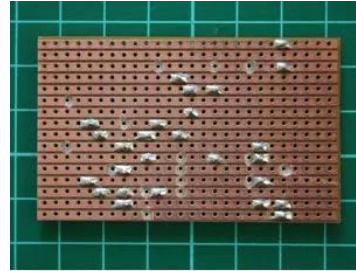
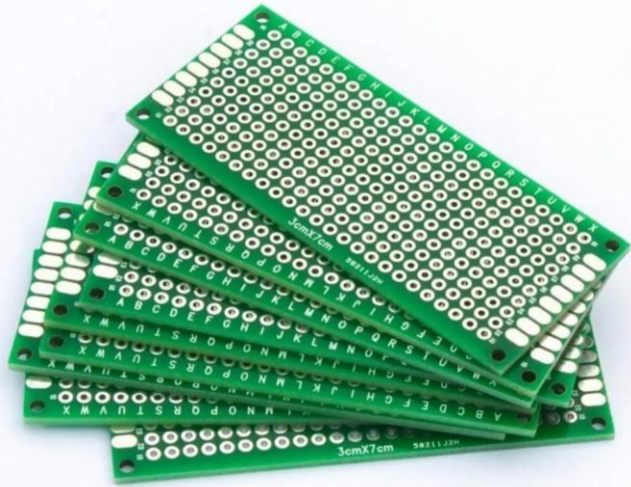
Schematic diagram



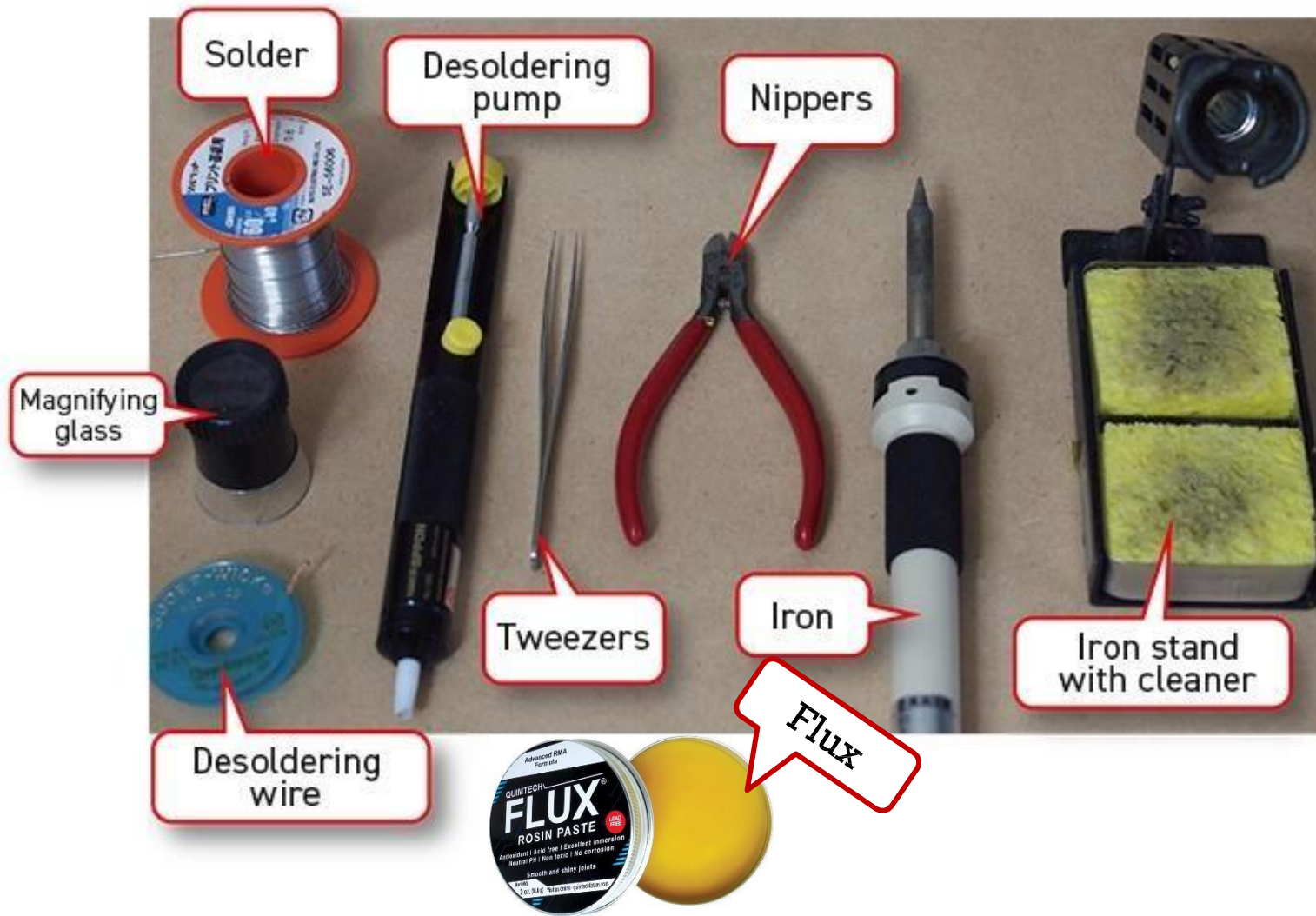
Real circuit using a solderless breadboard



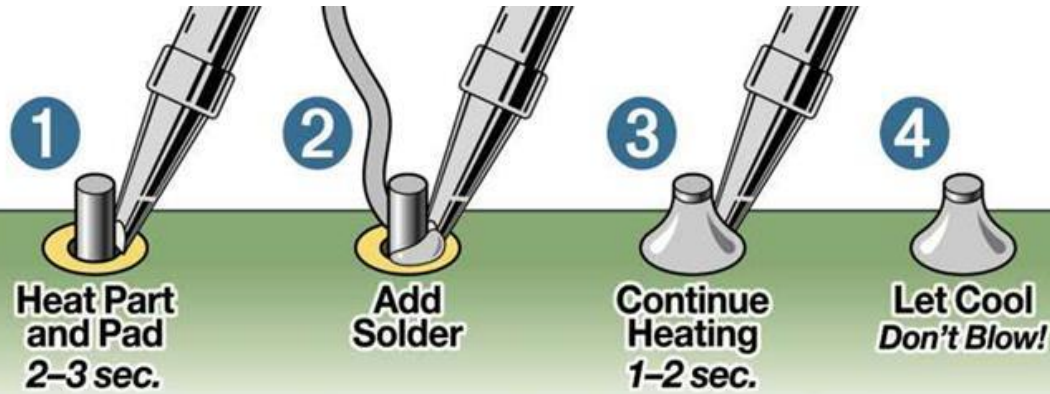
VERO BOARD & PRINTED CIRCUIT BOARD(PCB):



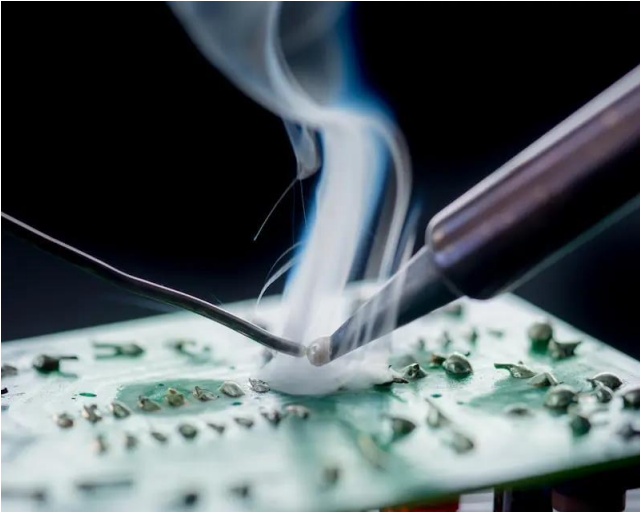
SOLDERING KIT:



PCB SOLDERING:

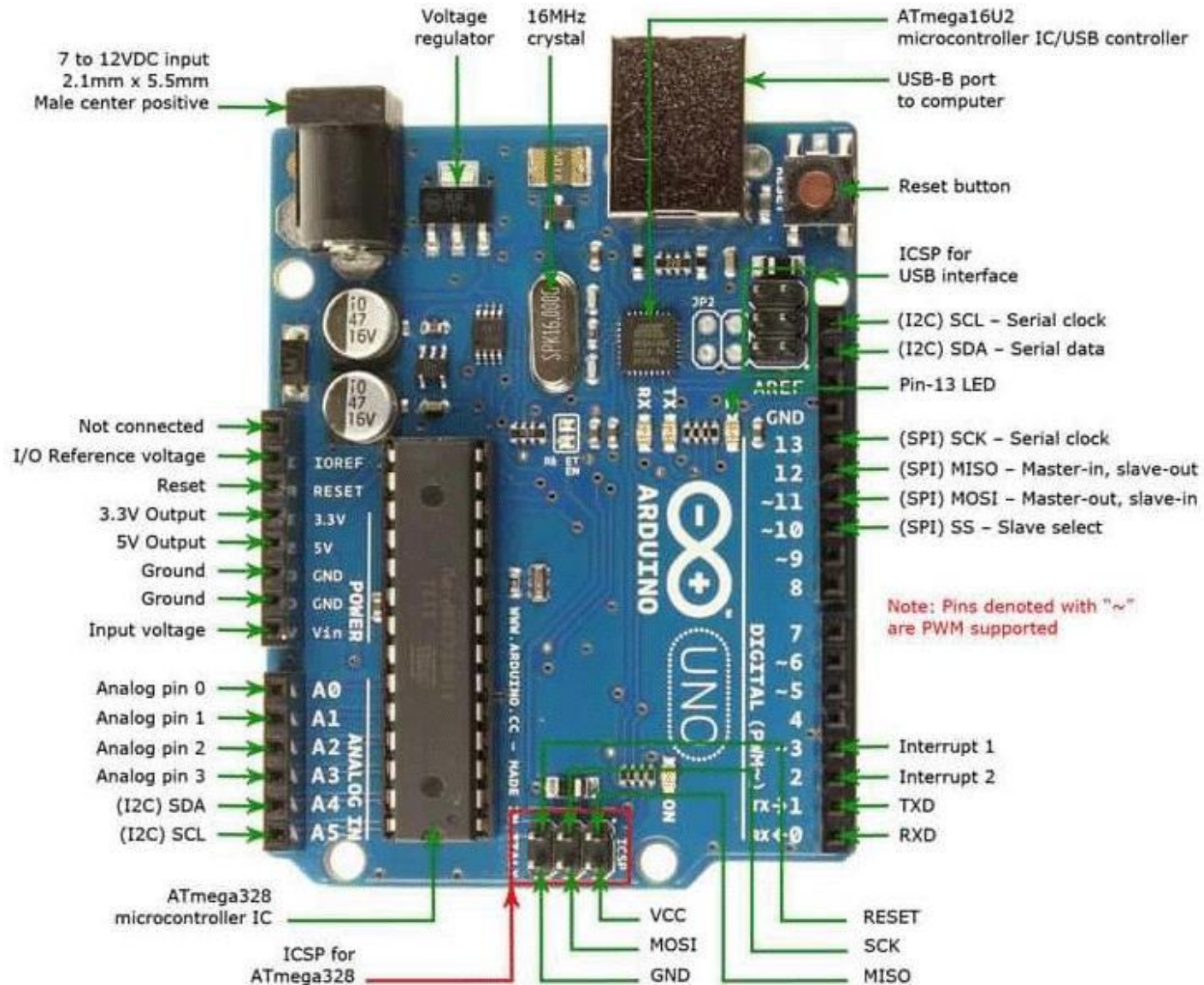


At start, and every few connections: clean tip of iron on damp sponge, apply thin layer of solder

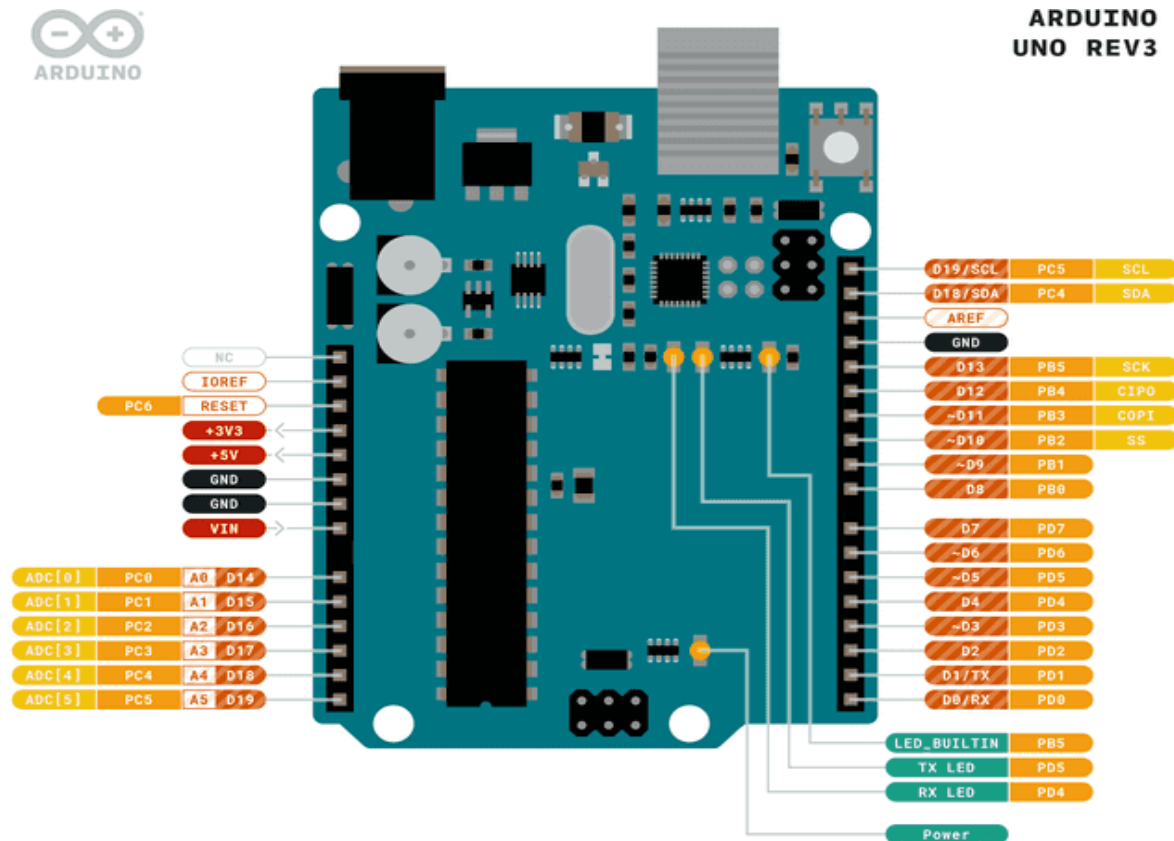


A collection of various electronic modules and sensors, including potentiometers, relays, LEDs, and various sensors, arranged in a grid. The components are mostly small circuit boards with different colors (black, red, blue) and various electronic components like resistors, capacitors, and integrated circuits. Some have headers for easy connection to breadboards or other modules.

ARDUINO UNO (SIMPLE PINOUT DIAGRAM):



ARDUINO UNO (FULL PINOUT DIAGRAM):



This work is licensed under the Creative Commons Attribution-ShareAlike 4.0 International License. To view a copy of this license, visit <http://creativecommons.org/licenses/by-sa/4.0/> or send a letter to Creative Commons, PO Box 1888, Mountain View, CA 94042, USA.





❖ PROTOTYPING SIMULATION PLATFORMS





TINKER CARD:

[HTTPS://WWW.TINKERCAD.COM/](https://www.tinkercad.com/)

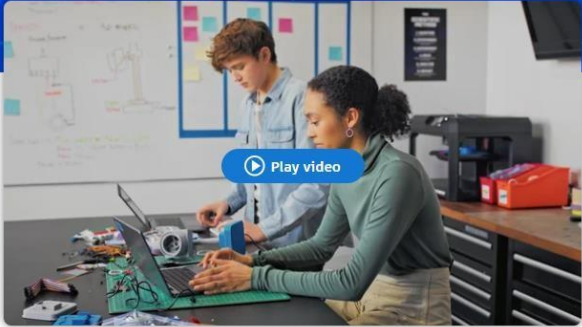
 **AUTODESK**
Tinkercad

Tinker ▾ Gallery Learn Teachers Resources ▾

Q Log In Sign Up

All you need is a 'what if...'





Tinkercad is a free web app for 3D design, electronics, and coding, trusted by over 75 million people around the world.





Tinker Card Login Page:



Welcome back

How do you use Tinkercad?

In school

Educators

Students with Class Code

Student accounts

On your own

Personal accounts

Don't have an account yet?

[Join Tinkercad](#)





In Circuit:



Tinker ^

Gallery

Learn

Teachers

Resources ^



3D Design

Start designing in 3D in minutes.



Circuits

Add light and movement to your designs.



Codeblocks

Write programs to bring your designs to life.



Sim Lab

Get things moving.



iPad App

Design on the go.



Autodesk Fusion

Level up your designs.

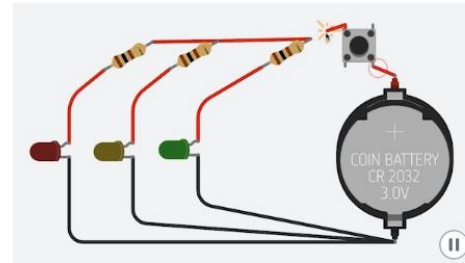


Autodesk Forma

Build your future now.

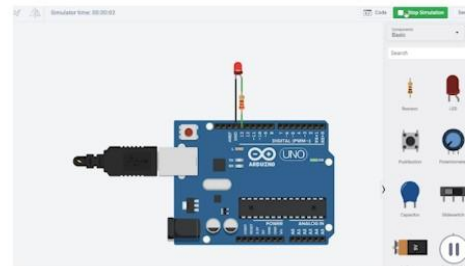
Design electronics

Place and wire electronic components (even a lemon) to create a virtual circuit from scratch, or use our starter circuits to explore and try things out. No additional hardware required.



Program with ease

Using our interactive circuit editor, you can explore, connect, and code virtual projects. Learning with Arduino or micro:bit? Open the code editor to program using Codeblocks.



Start simulating

Watch how components respond virtually before wiring your real-life circuit. Prototype your designs with 1:1 representations of Arduino, micro:bit, and other common components.





Contd.:



Ingenious Trug-Blad

All changes saved



Code

Start Simulation

Send To

Pushbutton

Name 1

Components
Basic

Search



Resistor



LED



Pushbutton



Potentiomet...



Capacitor



Slideswitch



9V Battery



Coin Cell 3V
Battery



1.5V Battery



Breadboard
Small



micro:bit



Arduino Uno
R3



Vibration
Motor



DC Motor



Micro Servo



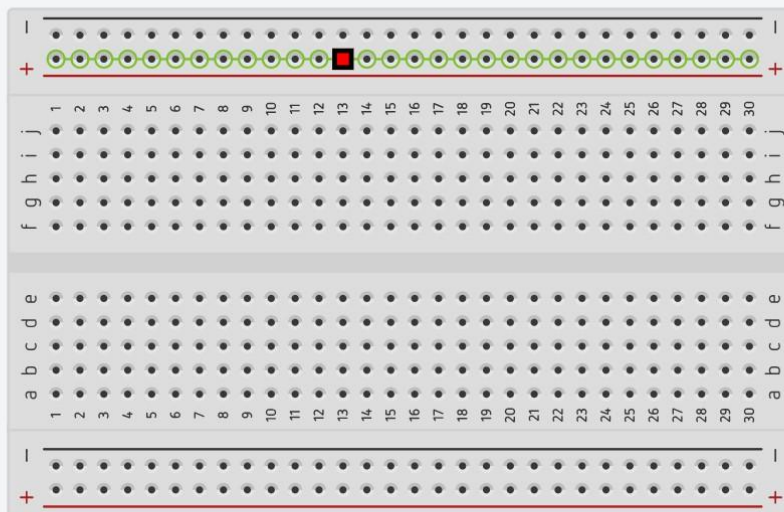
Hobby
Gearmotor



NPN
Transistor...



LED RGB



Start Simulating

Start Simulating

Let's learn how to test a circuit design with the simulator!

1/3

Start Simulating

Simulator time: 00:00:14

Code Stop Simulation Send To

Components Basic

Search

- Resistor
- LED
- Pushbutton
- Potentiomet...
- Capacitor
- Slideswitch
- 9V Battery
- Coin Cell 3V Battery
- 1.5V Battery
- Breadboard Small
- micro:bit
- Arduino Uno R3
- Vibration Motor
- DC Motor
- Micro Servo
- Hobby Gearmotor
- NPN Transistor...
- LED RGB

Next step





WOKWI: [HTTPS://WOKWI.COM/](https://wokwi.com/)



WOKWI

World's most advanced ESP32 simulator

Docs [SIGN IN](#)

[Discord Community](#) [LinkedIn Group](#) [Pricing](#)

Simulate with Wokwi Online



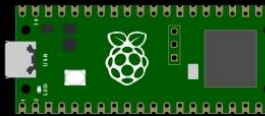
Arduino (Uno, Mega, Nano)



ESP32



STM32



Pi Pico





Wokwi Login Page:



WOKWi

World's most advanced ESP32 simulator

[Discord Community](#) [LinkedIn Group](#) [Pricing](#)

Docs



Online Arduino Simulator

A faster way to prototype Uno, Mega and Nano projects

Featured projects



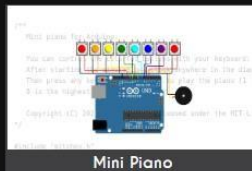
Simon Says Game



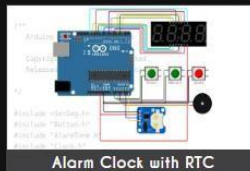
32 Servos Dancing



32x32 LED Matrix Tunnel



Mini Piano



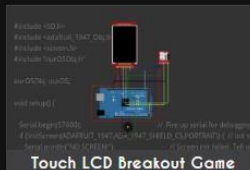
Alarm Clock with RTC



Electronic Safe



Nano Pong



Touch LCD Breakout Game



Dino Game



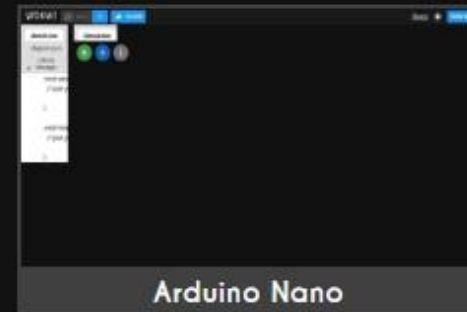
Start from Scratch:



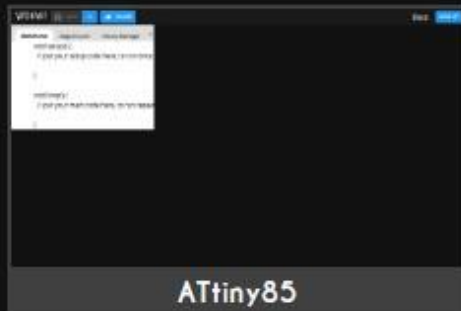
Arduino Uno



Arduino Mega



Arduino Nano



ATtiny85



Franzininho





Contd.:

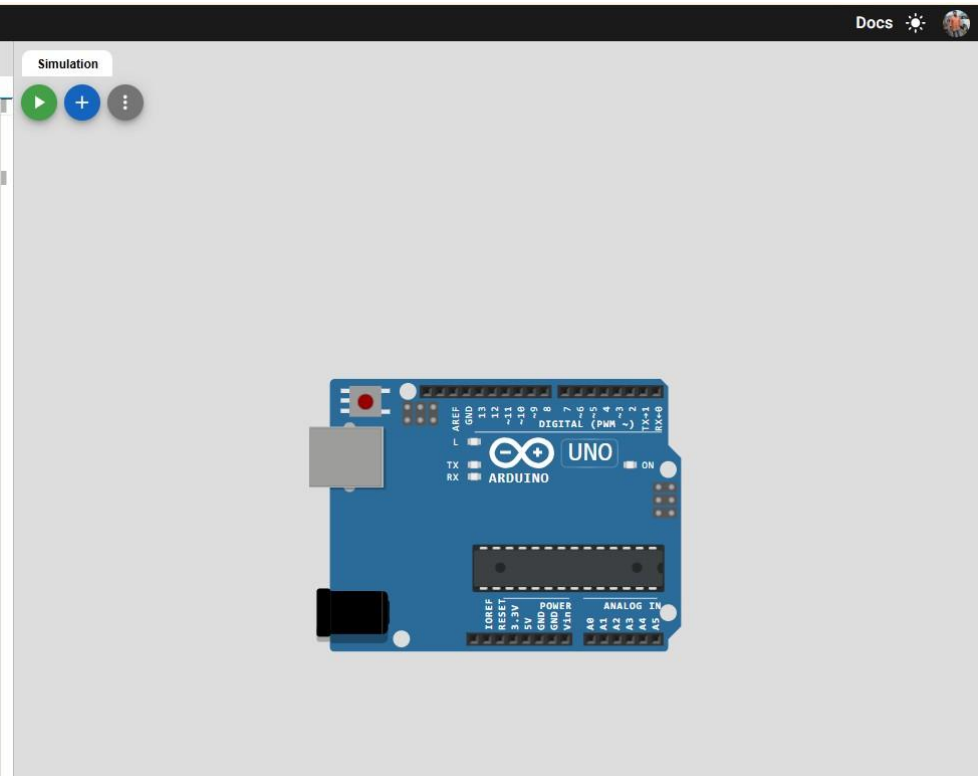


WOKWI

SAVE SHARE

sketch.ino diagram.json Library Manager

```
1 void setup() {  
2   // put your setup code here, to run once:  
3  
4 }  
5  
6 void loop() {  
7   // put your main code here, to run repeatedly:  
8  
9 }  
10
```



WOKWI SAVE SHARE

sketch.ino diagram.json Library Manager

```
1 void setup() {  
2   // put your setup code here, to run once:  
3 }  
4  
5  
6 void loop() {  
7   // put your main code here, to run repeatedly:  
8 }  
9  
10
```

Basic

- LED
- Pushbutton
- Pushbutton 6mm
- Resistor

Display

- RGB LED
- SSD1306 OLED display
- Grove SH1107 - OLED Display
- LCD 16x2
- LCD 16x2 (I2C)
- LCD 20x4
- LCD 20x4 (I2C)
- ILI9341 2.8" TFT-LCD display
- ILI9341 2.8" Touch Screen LCD
- LED Dot Matrix (MAX7219)

Docs





❖ ARDUINO INTEGRATED DEVELOPMENT ENVIRONMENT (IDE)





ARDUINO HOME:

[HTTPS://WWW.ARDUINO.CC/](https://www.arduino.cc/)



Solutions ▾

Products ▾

Community ▾

Documentation ▾

SHOP

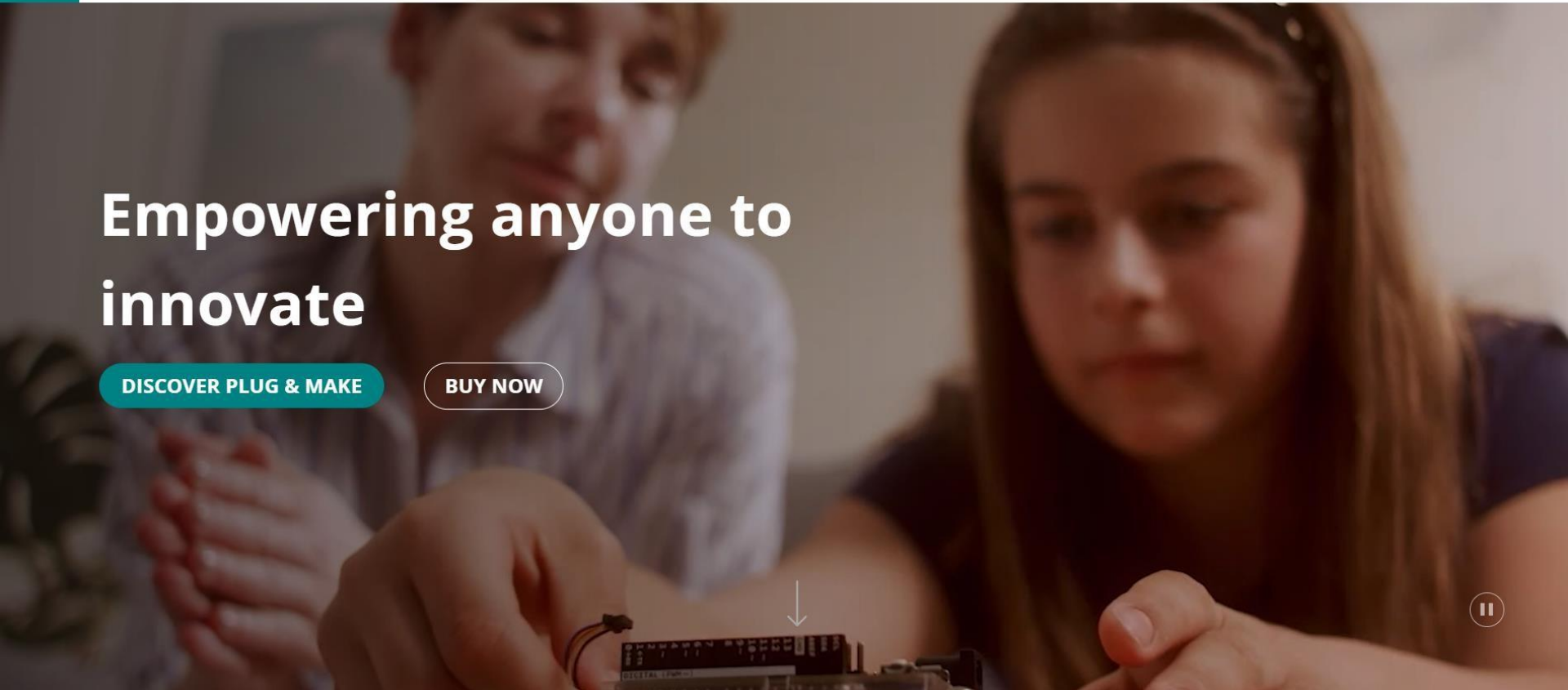


 Cloud

Empowering anyone to innovate

DISCOVER PLUG & MAKE

BUY NOW





Contd.:



For Professionals For Education For Makers

Products ^ Community v Documentation v SHOP | Cloud

HARDWARE

BOARDS, SOMs and SBCs

UNO

GIGA

Nano

Portenta

Nicla

ROBOTICS

Alvik

INDUSTRIAL AUTOMATION

Opta

KITS

Starter Kit

Plug & Make Kit

Science Kit

Portenta Proto Kit **NEW**

PLC Starter Kit

Sensor Kit

Student Kit

ALL HARDWARE →

SOFTWARE

CLOUD AND TOOLS

Arduino Cloud

Arduino Cloud Editor

Arduino CLI

Arduino IDE

PLC IDE

APP

IoT Remote

Science Journal

ALL SOFTWARE →



Arduino IDE

Discover all the features of our most popular programming tool



Arduino Cloud

Your next exciting journey to build and monitor your connected projects



Cloud Editor

Write code and upload sketches to any official Arduino board from your web browser



REGISTER YOUR ARDUINO PRODUCT

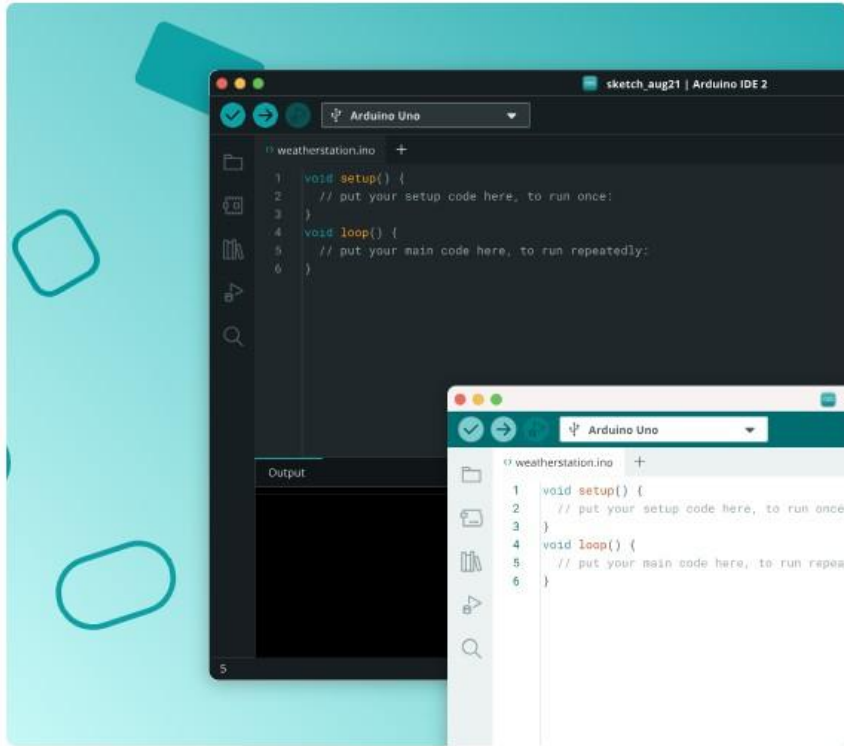
REGISTER



Software Download:

 arduino-ide_2.3.6_Windows_64bit	25-08-2025 12:21	Application	1,53,806 KB
---	------------------	-------------	-------------

Bring Your Projects to Life with Arduino Software



Arduino IDE 2.3.6

[Release notes](#)

The new major release of the Arduino IDE is faster and even more powerful! In addition to a more modern editor and a more responsive interface it features autocompletion, code navigation, and even a live debugger. For more details, check the [Arduino IDE 2.0 documentation](#).

Windows Win 10 or newer (64-bit)

DOWNLOAD

Windows Win 10 or newer (64-bit)

Windows MSI installer

Windows ZIP file

Linux AppImage (64-bit X86-64)

Linux ZIP file (64-bit X86-64)

macOS Intel 10.15 Catalina or newer (64-bit)

macOS Apple Silicon 11 Big Sur or newer (64-bit)



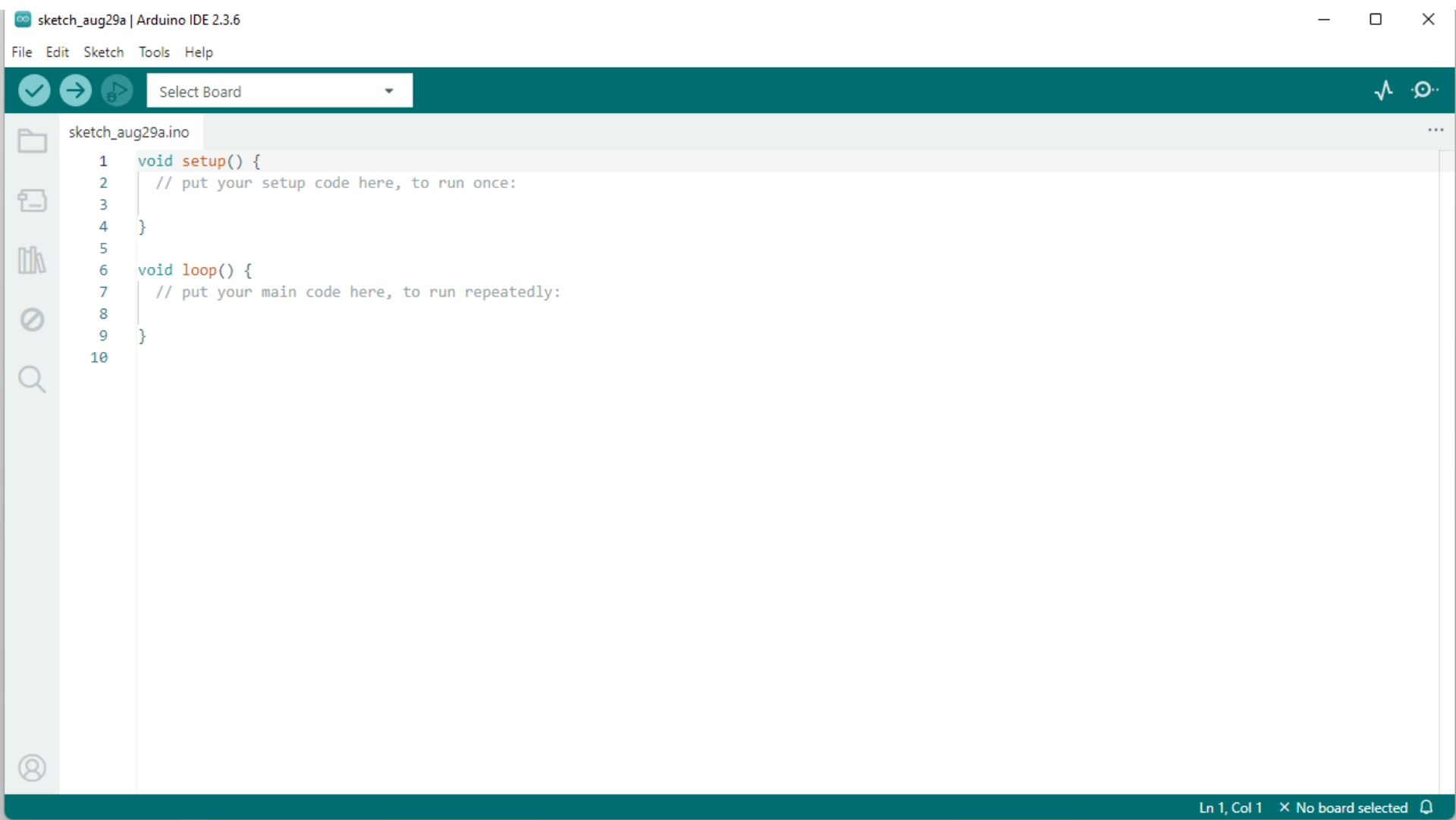
Legacy IDE (1.8.19)

Download a legacy version of the Arduino IDE.





Contd.:





Contd.:



sketch_aug29a | Arduino IDE 2.3.6

File Edit Sketch Tools Help

- New Sketch Ctrl+N
- New Cloud Sketch Alt+Ctrl+N
- Open... Ctrl+O
- Sketchbook >
- Examples >
 - Built-in examples
 - 01.Basics >
 - 02.Digital >
 - 03.Analog >
 - 04.Communication >
 - 05.Control >
 - 06.Sensors >
 - 07.Display >
 - 08.Strings >
 - 09.USB >
 - 10.StarterKit_BasicKit >
 - 11.ArduinoISP >
- Close Ctrl+W
- Save Ctrl+S
- Save As... Ctrl+Shift+S
- Preferences... Ctrl+Comma
- Advanced >
- Quit Ctrl+Q

Serial Monitor x

Not connected. Select a board to upload automatically.

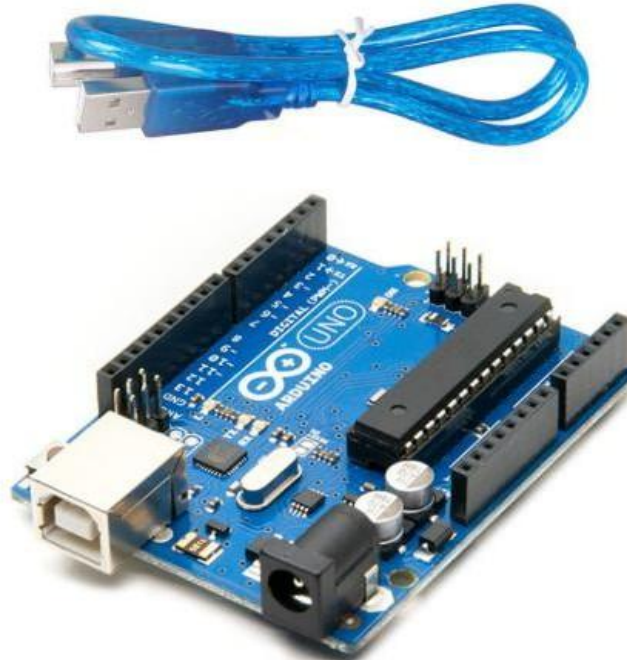
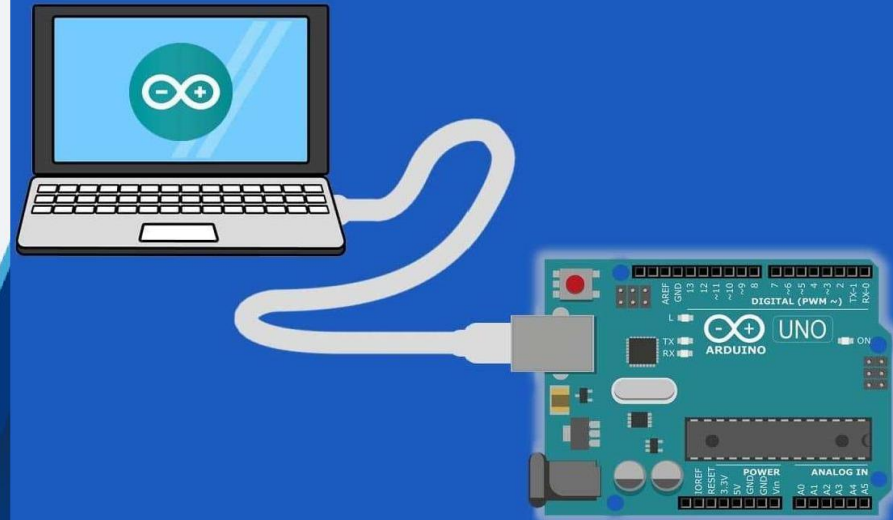
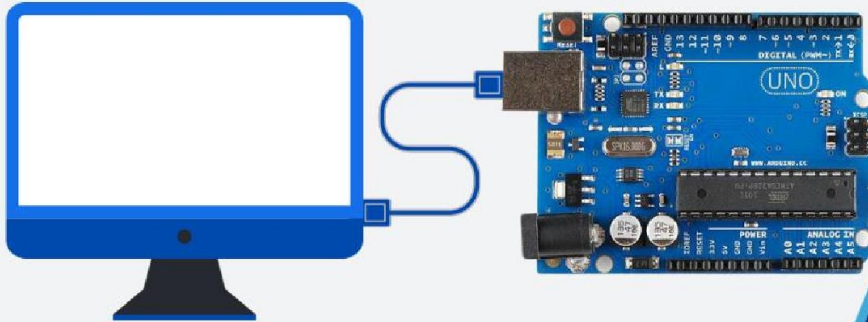
New Line

Couldn't open serial plotter

Ln 1, Col 1 x No board selected 1



Interfacing PC with Arduino



EXPERIMENTS

1. LED Blinking

